

Pumping lemma

1. $L = \{a^k \mid k \text{ is a prime number}\}$

Proof:

Let, L is a regular language. From the pumping lemma there exists a p such that every $w \in L$ such that $|w| \geq p$ can be represented as xyz with $|y| > 0$ and $|xy| \leq p$.

Let, m is a prime number where $m > p$.

We choose the string $w = a^m$ from the given language L . Now, according to the pumping lemma,

$$xy^iz \in L, i \geq 0 \text{ and } y = a^k, |y| = k, \text{ where } 0 < k \leq p$$

Now, if we take $i = 0$, $xy^iz = xz = a^{m-k}$

$$i = 2, xy^iz = xy^2z = xyyz = a^{m+k}$$

$$i = 3, xy^iz = xy^3z = a^{m+2k}$$

$$i = 4, xy^iz = xy^4z = a^{m+3k}$$

Now, considering $i = m + 1$, $xy^iz = xy^{m+1}z = a^{m+mk} = a^{m(1+k)}$

Here, $|a^{m(1+k)}| = m(1+k)$, which cannot be a prime number as $k > 0$ and it is divisible by both m and $(1+k)$. So, there is a contradiction. Hence, L is not a regular language.

2. $L = \{a^n \mid n \geq 0\}$

Proof:

Let, L is a regular language. From the pumping lemma there exists a p such that every $w \in L$ such that $|w| \geq p$ can be represented as xyz with $|y| > 0$ and $|xy| \leq p$.

We choose the string $w = a^p$ from the given language L . Now, according to the pumping lemma,

$$xy^iz \in L, i \geq 0 \text{ and } y = a^k, |y| = k, \text{ where } 0 < k \leq p$$

Taking $i = 0$, we get $xy^iz = xy^0z = xz = a^{p!-k}$

Here, $|a^{(p!-k)}| = p! - k$. But this length cannot be a factorial of any integer as we know that $(p!-k) < (p-1)!$, for $0 < k \leq p$. So, there is a contradiction. Hence, L is not a regular language.

3. $L = \{a^n b^l \mid n \neq l\}$

Proof:

Let, L is a regular language. From the pumping lemma there exists a p such that every $w \in L$ such that $|w| \geq p$ can be represented as xyz with $|y| > 0$ and $|xy| \leq p$.

We choose the string $w = a^{p!} b^{(p+1)!}$ from the given language L .

(Note: You need to choose a string either with factorials or with LCM of the numbers from 1 to p for b . Or else the proof will be incorrect)

Now, according to the pumping lemma,

$$xy^iz \in L, i \geq 0 \text{ and, } y = a^k, |y| = k, \text{ where } 0 < k \leq p$$

(This proof is done differently. Here, we will find if there is any integer value of i that exists for which the length of a 's and b 's will be the same. If we can prove that the integer exists, then the contradiction will be created.)

For the given string w to be a part of the language L , an integer i shouldn't exist for which $n = l$. If $n = l$,

$$p! + (i - 1)k = (p + 1)!$$

$$\text{Or, } (i - 1)k = (p + 1)! - p!$$

$$\text{Or, } (i - 1)k = (p+1).p! - p! = p!(p+1-1) = p!.p$$

$$\text{Or, } (i - 1) = (p!.p) / k$$

$$\text{Or, } i = (p!.p)/k + 1$$

Here, i will be an integer value if $p!.p$ is completely divisible by k . This is possible as k is an integer where $0 < k \leq p$, so it can divide $p!$ and produce an integer value. Since i is an integer, then it is possible for the value of n and l to be equal, which is not in the language.

So, since contradiction is created, L is not a regular language.

An alternative and easier solution: The complement of a regular language should also be a regular language. Sp, if $\bar{L}$ is regular, L will also be regular and vice versa. $\bar{L} = \{a^n b^l \mid n = l\}$. Now you need to prove that $\bar{L}$ is not regular. Only after proving $\bar{L}$ is not regular, conclude that L is also not regular.

4. $L = \{a^k b^n c^l \mid k \neq n + l\}$

Proof: Same logic as before